Electronics Workshop 2 - 2026

Course Project 2

Team Details

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Project

Create a MEMS microphone preamplifier with automatic gain control (AGC) and anti-aliasing filter for always-on voice wake-word detection in smart home devices.

Proposed Solution

A low power circuit using a preamplifier circuit along with the MEMS microphone interface, succeeded by an automatic gain control (AGC) stage, a gain circuit, an Anti Aliasing Filter and an Active Band Pass Filter, connected to an ADC and the ESP, interfaced with a Language Model aboard a Raspberry Pi 4 for **wake word identification** and **querying**.

Stages

- Stage 1: MEMS Interface
- Stage 2: Pre-Amp Stage
- Stage 3: Automatic Gain Control
- Stage 4: Gain Stage
- Stage 5: Anti-Aliasing Filter
- Stage 6: Active Band Pass Filter
- Stage 7: ADC
- Stage 8: ESP32
- Stage 9: Raspberry Pi & TinyML model

Stage 1: MEMS Interface

Component: SPW2430 MEMS Microphone & Breakout Board

- Detects acoustic pressure and converts it to an analog voltage signal
- Small form factor, high sensitivity – ideal for voice recognition
- Requires 3.3 V DC at V_{in}; AC-coupled output fed to Pre-Amp
- Typical maximum output: $\approx 100\,\mathrm{mV}_{pp}$

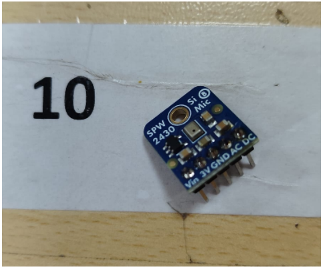


Figure: SPW2430 MEMS Microphone Module

Stage 2: Pre-Amplifier – Theory

Topology: Common-Emitter Differential Amplifier (NPN BJT pair)

- High input impedance, low output impedance, good noise rejection
- Amplifies differential signal; rejects common-mode noise (CMRR)
- **Single Input, Unbalanced Output** configuration used

Key gain expressions:

$$A_d = -\frac{g_m R_C}{2}, \qquad A_c = -\frac{\alpha R_C}{r_e + 2R_{EE}}$$

$$\mathrm{CMRR} = 20\log\left(\frac{A_d}{A_c}\right) \approx 20\log(g_m R_{EE})$$

Design target: Single-ended differential gain $A_d \approx 10$

Component values:

- $R_C = 1.5\,\mathrm{k}\Omega$ (collector load)
- $R_{EE} = 3.3\,\mathrm{k}\Omega$ (tail resistor, $-5\,\mathrm{V}$ rail)
- BJT: BC547B ($\beta = 200$)

Operating point:

$$I_C \approx 0.65\,\mathrm{mA}, \quad g_m \approx 26\,\mathrm{mA/V}$$

$$A_d \approx 19.5, \quad A_c \approx 0.23$$

$$\mathrm{CMRR} \approx 38.7\,\mathrm{dB}$$

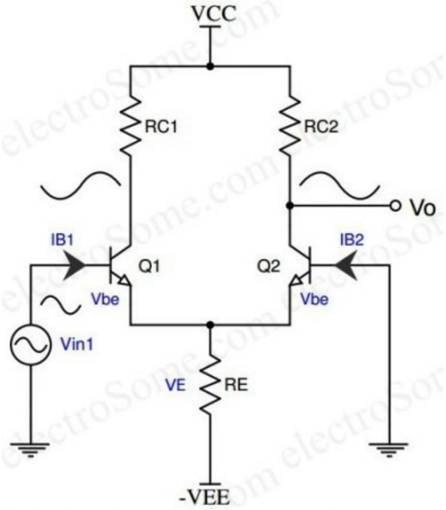


Figure: Pre-Amp Circuit Diagram

Stage 3: Automatic Gain Control (AGC)

Components:

- BF245B N-Channel JFET – Variable gain element
- UA741 Op-Amp – Inverting amplifier
- BC557C PNP BJT – Envelope detector
- $R_4 = 100\,\mathrm{k}\Omega,\ R_{17} = 10\,\mathrm{k}\Omega,\ C_2 = 10\,\mu\mathrm{F},\ R_5 = 156\,\mathrm{k}\Omega$

Three functional blocks:

1. **Variable Gain Amplifier** – JFET as gate-controlled resistor
2. **Envelope Detector** – BC557C tracks signal peak into C_2
3. **Control Feedback** – V_C drives JFET gate (negative feedback)

AGC Theory: Variable Gain & Feedback

JFET channel resistance (ohmic region):

$$r_{DS}(V_{GS}) = \frac{r_{DS0}}{1 - V_{GS}/V_P}, \quad r_{DS0} = 250\,\Omega,\ V_P \approx -2\,\mathrm{V}$$

Inverting amplifier gain:

$$A_v(V_{GS}) = -\frac{R_4}{R_{17} + r_{DS}(V_{GS})} = -\frac{100\,\mathrm{k}}{10\,\mathrm{k} + r_{DS}}$$

Range: 0 to ≈ -9.76 (0 to $\approx 19.8\,\mathrm{dB}$)

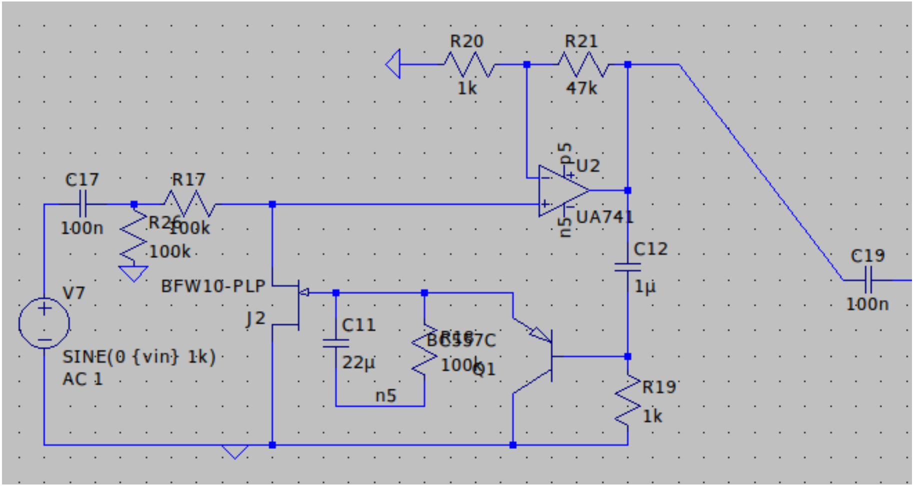
AGC time constants:

$$\tau_{\text{attack}} \ll 1\,\mathrm{ms}, \qquad \tau_{\text{release}} = R_5 \cdot C_2 = 156\,\mathrm{k\Omega} \times 10\,\mu\mathrm{F} = \mathbf{1.56}\,\mathrm{s}$$

Slow release prevents "pumping" artefacts during speech pauses.

Output coupling: $f_c = 1/(2\pi \cdot 4.7\,\mathrm{k} \cdot 1\,\mu) \approx 33.9\,\mathrm{Hz}$

AGC LTspice Simulation Circuit



Topology: Common-Emitter amplifier with voltage-divider bias and split emitter resistors along with emitter degeneration.
We have designed for a voltage gain of 2.
The voltage gain]

Purpose: Low-pass filter before ADC to prevent aliasing (Nyquist–Shannon theorem)

Topology: 4th-order Butterworth – two cascaded 2nd-order Sallen-Key stages

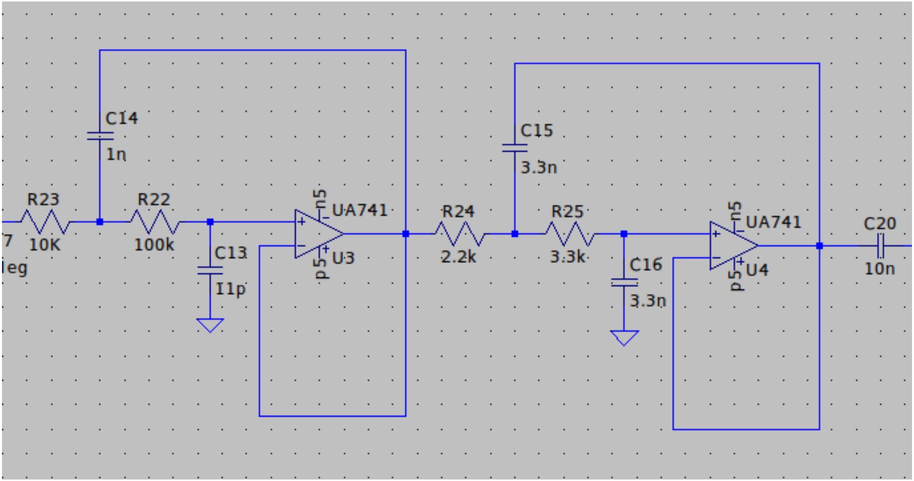
- Maximally flat passband – minimal distortion over voice band
- Rolloff: $-80\,\mathrm{dB/decade}$ beyond cutoff
- Cutoff set to $10\,\mathrm{kHz}$ (half the $20\,\mathrm{kHz}$ sampling rate)

Sallen-Key transfer function:

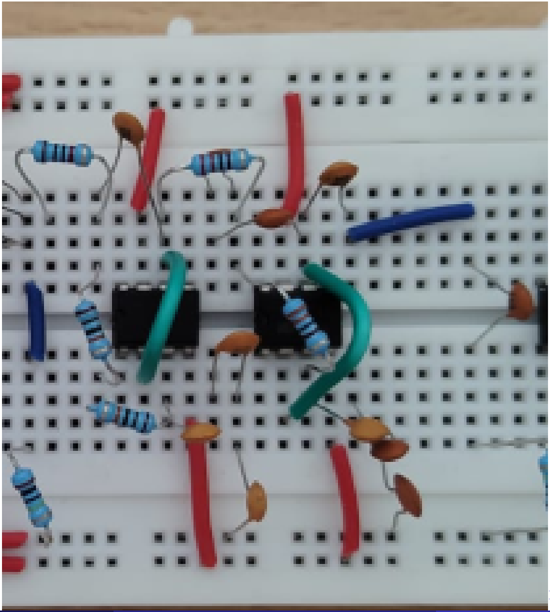
$$H(s) = \frac{1}{s^2 R_1 R_2 C_1 C_2 + sC_2(R_1 + R_2) + 1}$$

$$\omega_c = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \qquad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{C_2(R_1 + R_2)}$$

AAF LTspice Simulation Circuit



AAF Hardware Circuit



Purpose: Provide high-pass filtering (DC blocking) and additional low-pass filtering with amplification in a single stage.

Topology: Passive HPF $\rightarrow$ Non-inverting Op-Amp $\rightarrow$ Passive LPF

Overall transfer function:

$$H(s) = \left(1 + \frac{R_4}{R_3}\right) \cdot \frac{sR_1 C_1}{(1 + sR_1 C_1)(1 + sR_2 C_2)}$$

Designed cutoffs:

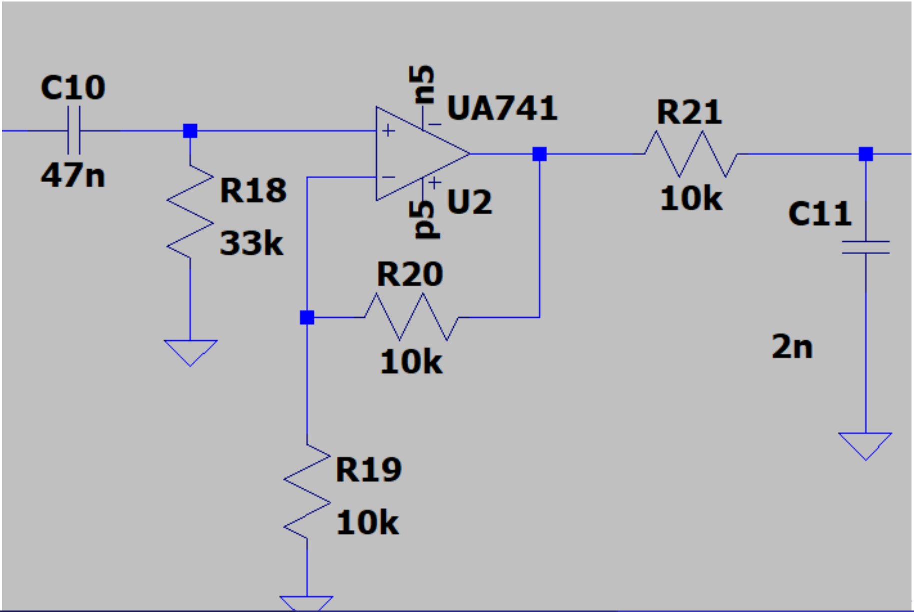
$$f_{HP} = \frac{1}{2\pi \cdot 56\,\mathrm{k} \cdot 47\,\mathrm{n}} \approx 60.5\,\mathrm{Hz}$$

$$f_{LP} = \frac{1}{2\pi \cdot 10\,\mathrm{k} \cdot 2\,\mathrm{n}} \approx 7.96\,\mathrm{kHz}$$

Amplifier gain:

$$A_v = 1 + \frac{10\,\mathrm{k}}{10\,\mathrm{k}} = \mathbf{2}$$

BPF LTspice Simulation Circuit



Input Protection: Schottky Diode Clamp (BAT54S)

Purpose: Protect ESP32 ADC from voltages outside 0–3.3 V

Topology: Rail-to-rail clamp ($V_F \approx 0.25\,\mathrm{V}$ for BAT54S)

$$V_{high} = 3.3 + 0.25 = \mathbf{3.55}\,\mathrm{V}, \qquad V_{low} = 0 - 0.25 = -\mathbf{0.25}\,\mathrm{V}$$

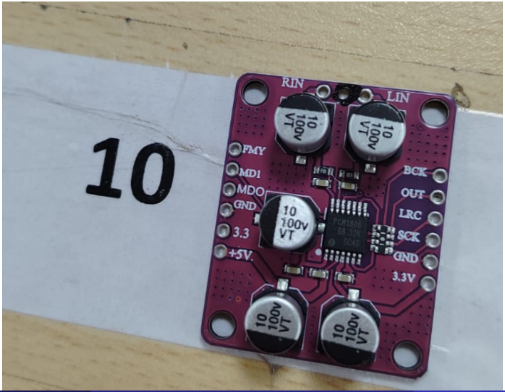
Both within the ESP32 absolute maximum of 3.6 V.

In normal operation both diodes are reverse-biased and do not affect the signal. A standard silicon diode ($V_F \approx 0.65\,\mathrm{V}$) would clamp at 3.95 V – above the limit.

Stage 7: Analog-to-Digital Converter (PCM1808)

Component: PCM1808 Stereo ADC (Texas Instruments)

- Single-ended analog input, supports sample rates up to 96 kHz
- Operates in **slave mode** – I^{2}S master clock provided by ESP32
- System clock: $256f_S$, $384f_S$, or $512f_S$ (auto-detected)
- 24-bit audio data, left-justified within 32-bit I^{2}S frame



Stage 8: ESP32 – I²S Master & Serial Bridge

Role: Drive the PCM1808 ADC and stream audio to Raspberry Pi over UART

- $\mathrm{I}^2\mathrm{S}$ configured at **16 kHz**, 32-bit samples, mono (right channel)
- APLL enabled for low-jitter master clock on GPIO 3 (MCLK)
- BCK / LRCK / DATA on GPIOs 26 / 25 / 22
- Each 32-bit sample right-shifted by 16 bits $\rightarrow$ 16-bit PCM
- **Framed UART transmission at 460800 baud**:
  - 4-byte sync header: `0xDE 0xAD 0xBE 0xEF`
  - 256 bytes audio payload (128 samples $\times$ 2 bytes)
- 8 DMA buffers $\times$ 512 bytes to prevent data loss

Stage 9: Raspberry Pi 4 – Wake Word Detection

Role: Host processor running the full wake-word pipeline

1. **Frame Sync:** Scan byte stream for `0xDE 0xAD 0xBE 0xEF` header
2. **DSP Chain:**
   - 4th-order Butterworth BPF: 80–7500 Hz (SOS, 64-bit float)
   - Linear gain $5\times$, hard-clipped to 16-bit range
3. **Wake-Word Inference:**
   - *openWakeWord* engine, pre-trained *Alexa* model
   - Input: 1280 samples (80 ms at 16 kHz)
   - Detection threshold: 0.1; debounce: 1.5 s

Final circuit demonstrated wake word detection with decent confidence scores in both noisy and non-noisy environments.

Echo Dot (3rd Generation) Power Consumption

Mode	**Power Consumption (in Watts)**
Off Mode* (Low Power Mode)	0.16
Networked Standby** **(Automatic Sleep Mode, Microphones ON)**	1.40
Networked Standby **(Automatic Sleep Mode, Microphones OFF)**	2.04

Figure: Power consumption metrics of the Amazon Alexa/Echo model with voice wake word detection Click here

Our Alexa!

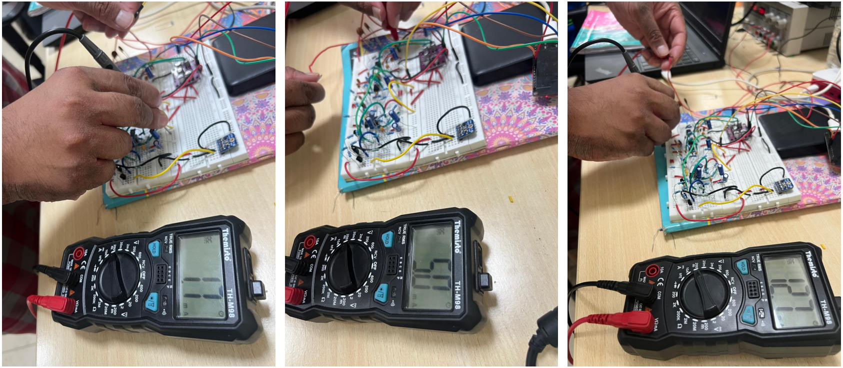


Figure: Pictures taken while measuring power consumed by our circuit

$$P_{\text{total}} = (3.3\,\text{V} \times 1.7\,\text{mA}) + (5\,\text{V} \times 12.7\,\text{mA}) + (5\,\text{V} \times 5\,\text{mA})$$

$$P_{\text{total}} = 5.61\,\text{mW} + 63.5\,\text{mW} + 25\,\text{mW} = \mathbf{94.11}\,\text{mW}$$

Conclusion

- Successfully designed and implemented a complete analog front-end for always-on wake word detection
- **Signal chain:** MEMS mic $\rightarrow$ Pre-Amp $\rightarrow$ AGC $\rightarrow$ Gain $\rightarrow$ AAF $\rightarrow$ BPF $\rightarrow$ ADC $\rightarrow$ ESP32 $\rightarrow$ Raspberry Pi
- AGC maintains constant output across varying speaker distances
- 4th-order Butterworth AAF provides $-80\,\mathrm{dB/decade}$ attenuation
- **Total power consumption: 94.11 mW** (comparable to commercial devices)
- Wake word "Alexa" detected with good confidence in noisy and quiet environments